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# Inventory Stock Analysis System - Phase 3

products = ["Laptop", "Mouse", "Keyboard", "Monitor", "Headphones"]
stock = [10, 25, 15, 12, 8]
minimum = [5, 10, 5, 5, 5]
price = [50000, 500, 1200, 8000, 1500]

while True:
    print("\n--- INVENTORY STOCK ANALYSIS ---")
    print("1. View Products")
    print("2. Add Product")
    print("3. Update Stock")
    print("4. Low Stock Alert")
    print("5. Stock Summary")
    print("6. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        print("\nProduct\t\tStock\tPrice")
        for i in range(len(products)):
            print(products[i], "\t", stock[i], "\t", price[i])

    elif choice == "2":
        name = input("Enter product name: ")

        if name.lower() in [p.lower() for p in products]:
            print("Product already exists!")
        else:
            try:
                qty = int(input("Enter quantity: "))
                cost = float(input("Enter price: "))
                min_qty = int(input("Enter minimum stock: "))

                if qty < 0 or cost < 0 or min_qty < 0:
                    print("Values cannot be negative!")
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            else:
                products.append(name)
                stock.append(qty)
                price.append(cost)
                minimum.append(min_qty)
                print("Product added successfully!")
        except ValueError:
            print("Please enter valid numbers.")

elif choice == "3":
    name = input("Enter product name: ")
    found = False

    for i in range(len(products)):
        if products[i].lower() == name.lower():
            try:
                qty = int(input("Enter new stock quantity: "))
                if qty >= 0:
                    stock[i] = qty
                    print("Stock updated successfully!")
                else:
                    print("Quantity cannot be negative.")
            except ValueError:
                print("Please enter a valid quantity.")
            found = True
            break

    if not found:
        print("Product not found!")

elif choice == "4":
    print("\nLow Stock Products:")
    found = False

    for i in range(len(products)):
        if stock[i] <= minimum[i]:
            print(products[i], "- Stock:", stock[i])
            found = True
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    if not found:
        print("No low-stock products.")

elif choice == "5":
    total_products = len(products)
    total_stock = sum(stock)
    total_value = 0

    for i in range(len(products)):
        total_value += stock[i] * price[i]

    print("\n--- STOCK SUMMARY ---")
    print("Total Products:", total_products)
    print("Total Stock Units:", total_stock)
    print("Total Inventory Value: Rs.", total_value)

elif choice == "6":
    print("Thank you for using the system!")
    break

else:
    print("Invalid choice! Please try again.")
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